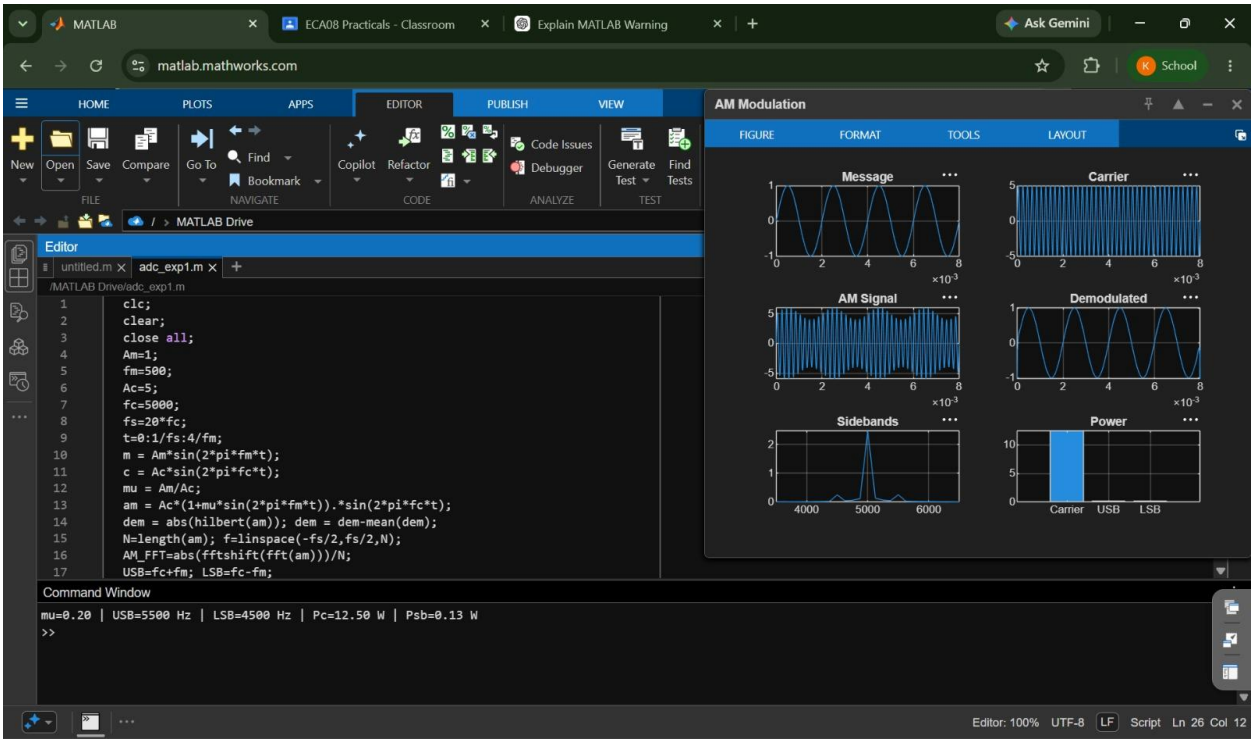
Experiment


MATLAB
ECA08 Practicals - Classroom
Explain MATLAB Warning
Ask Gemini
matlab.mathworks.com
School
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
New
Open
Save
Compare
Go To
Find
Bookmark
Copilot
Refactor
Code Issues
Debugger
Generate Test
Find Tests
FILE
NAVIGATE
CODE
ANALYZE
TEST
MATLAB Drive
Editor
untitled.m
adc_exp1.m
/MATLAB Drive/adc_exp1.m
clc;
clear;
close all;
Am=1;
fm=500;
Ac=5;
fc=5000;
fs=20*fc;
t=0:1/fs:4/fm;
m = Am*sin(2*pi*fm*t);
c = Ac*sin(2*pi*fc*t);
mu = Am/Ac;
am = Ac*(1+mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);
dem = abs(hilbert(am)); dem = dem-mean(dem);
N=length(am); f=linspace(-fs/2,fs/2,N);
AM_FFT=abs(fftshift(fft(am)))/N;
USB=fc+fm; LSB=fc-fm;
Command Window
mu=0.20 | USB=5500 Hz | LSB=4500 Hz | Pc=12.50 W | Psb=0.13 W
>>
AM Modulation
FIGURE
FORMAT
TOOLS
LAYOUT
Message
Carrier
AM Signal
Demodulated
Sidebands
Power
Carrier
USB
LSB
Editor: 100%
UTF-8
LF
Script
Ln 26 Col 12